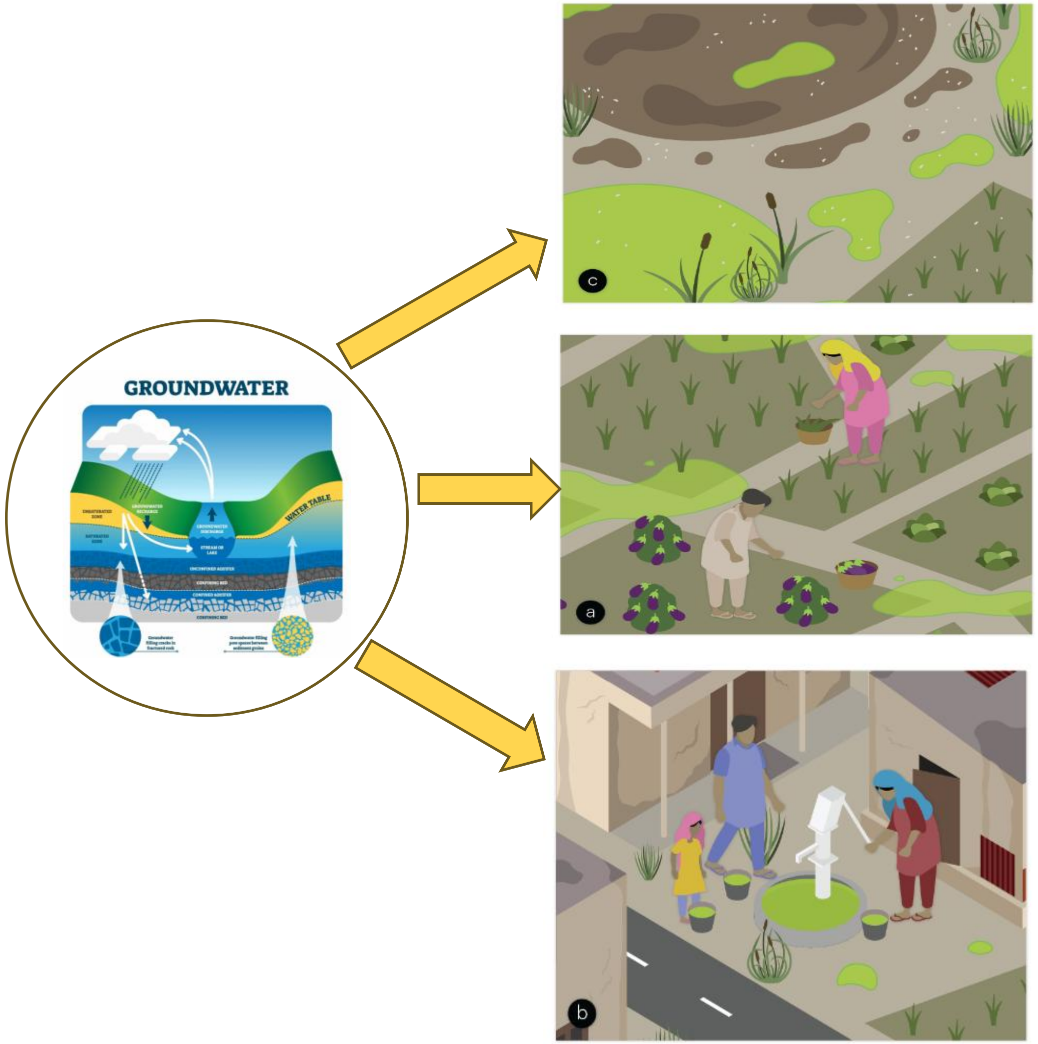
Chromium Contaminated Soil-Groundwater Systems at COPR Site in Rania and Khan Chandpur Villages, India

Introduction

- Contamination in the soil-groundwater system poses a **<u>serious problem in villages</u>**, as most of drinking water comes from groundwater sources.

- Approximately <u>600+ million people are at a high risk</u> due to contaminated soil and groundwater resources in India (Nitti Ayog, 2018).



"Managing soil and groundwater quality is ***crucial for rural development and livelihoods****."*

Soil-Groundwater Contaminants

Types of Contamination

Organic contaminants

Heavy Metals contaminants

Cr, Cd, Pb etc

As, Fe, Zn, etc

Chromium

Sources

Geogenic

Chromite Ore Processing Residual (COPR)

Properties

Trivalent Cr [Cr(III)] ~70%

Less Mobile
Less Toxic

Hexavalent Cr [Cr(VI)] ~30%

High Mobile
High Toxic

"Cr is a major concern for soil and groundwater quality and thus for the community health."

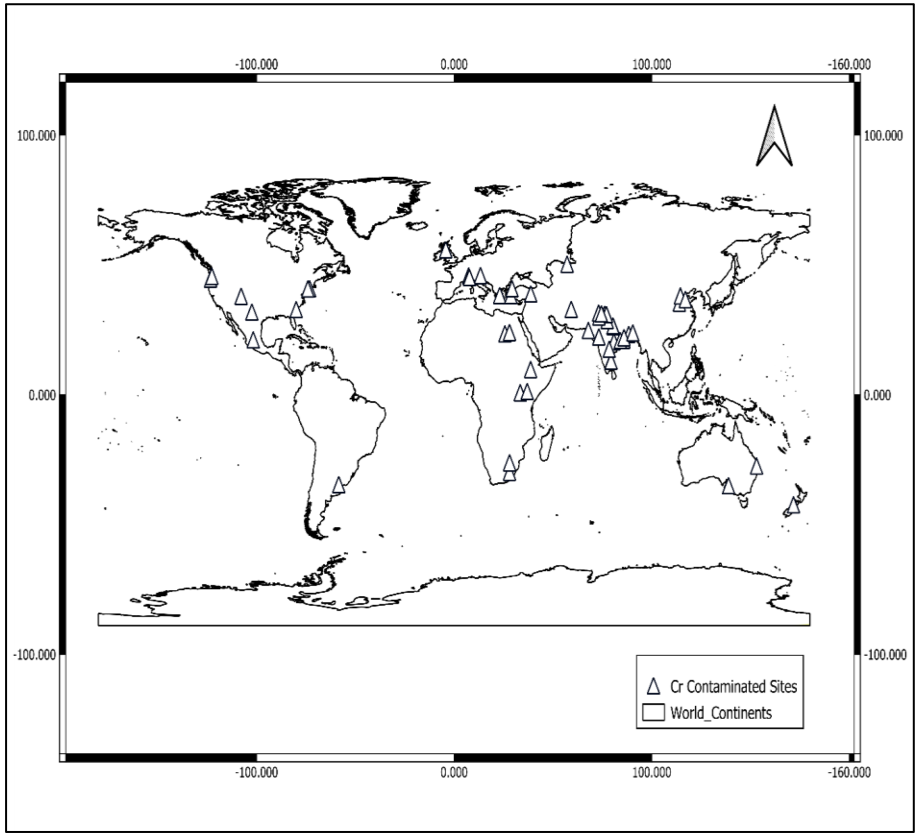
Cr Contaminated Groundwater for drinking

Health Risk

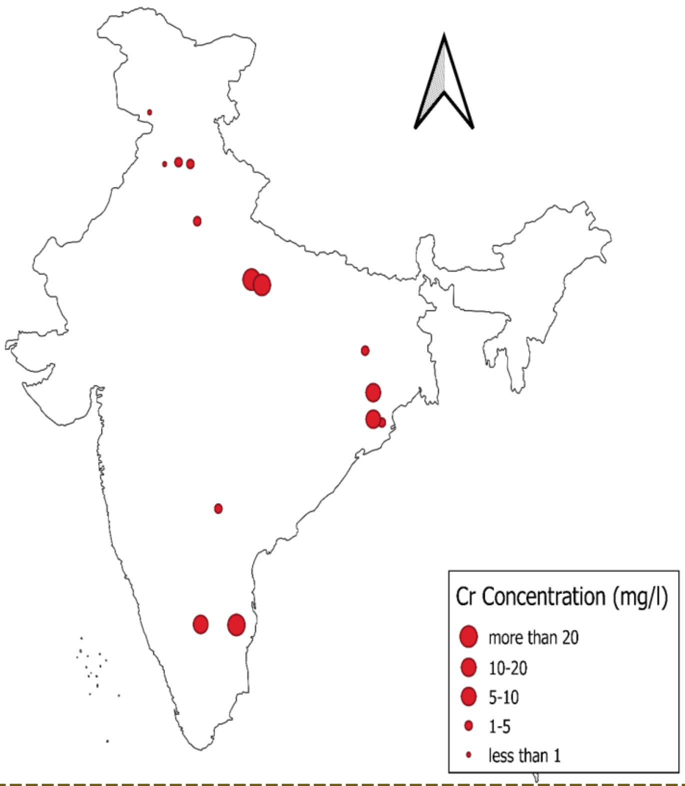
High Cancer Risk

Cr-Contaminated Sites

[Cr-Contaminated sites across the globe]



[Cr-Contaminated sites in India]

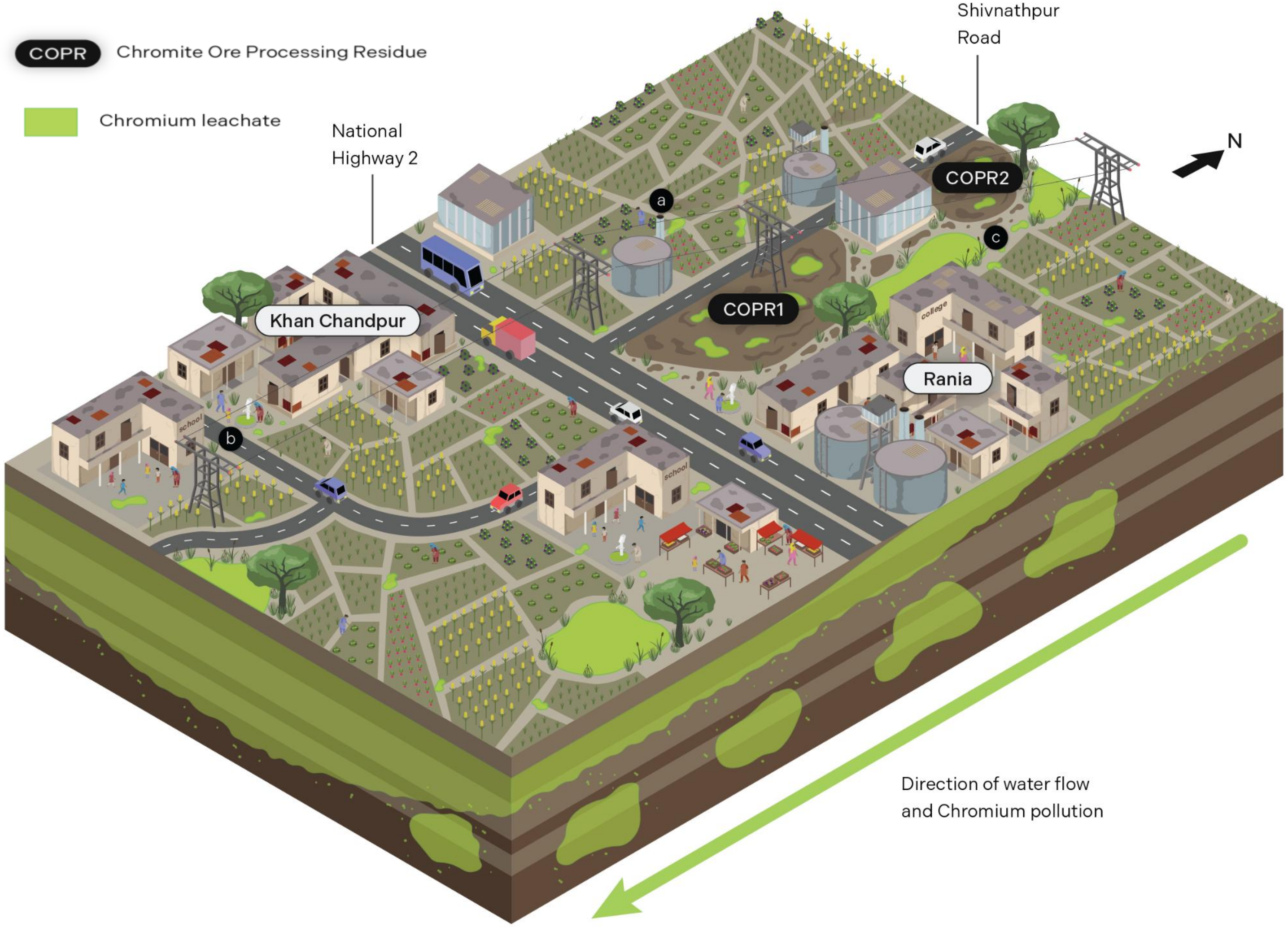


Site	Latitude	Longitude	Cr (mg/l)
Chhiwali	26.193	80.546	115
Rania	26.402	80.048	46
Ranipet	12.948	79.319	40.52
Bangalore	12.971	77.594	16.2
Nuggihalli	13.054	76.493	12
Sholan	30.900	77.096	1.07
Jajpur	20.834	86.332	2.48
Sukinda	20.964	85.917	11.35
Ramagarh	23.634	85.526	1.64
Chhanai	13.084	80.270	0.99
Jajmau	26.425	80.402	15
Papum Pare	27.224	93.500	0.18
Rupnagar	30.9752	76.527	1.5
Balangar	17.478	78.446	3.5
Odisha	22.000		7.2
Ghaziabad	28.676		1.5

112+

"No. of contaminated sites in India are increasing and ***most of such sites are located in rural*** *or peri-urban areas."*

Cr-Contaminated Rania & Khan Chandpur Villages



<u>2022-Onwards:</u> The CRDT Team has undertaken this site to study

<u>2021-Onwards:</u> Hon'ble NGT order to demonstrate remediation (No action taken yet by any agency due to lack of knowledge)

<u>2019:</u> Hon'ble NGT ordered to stop dumping in Rania

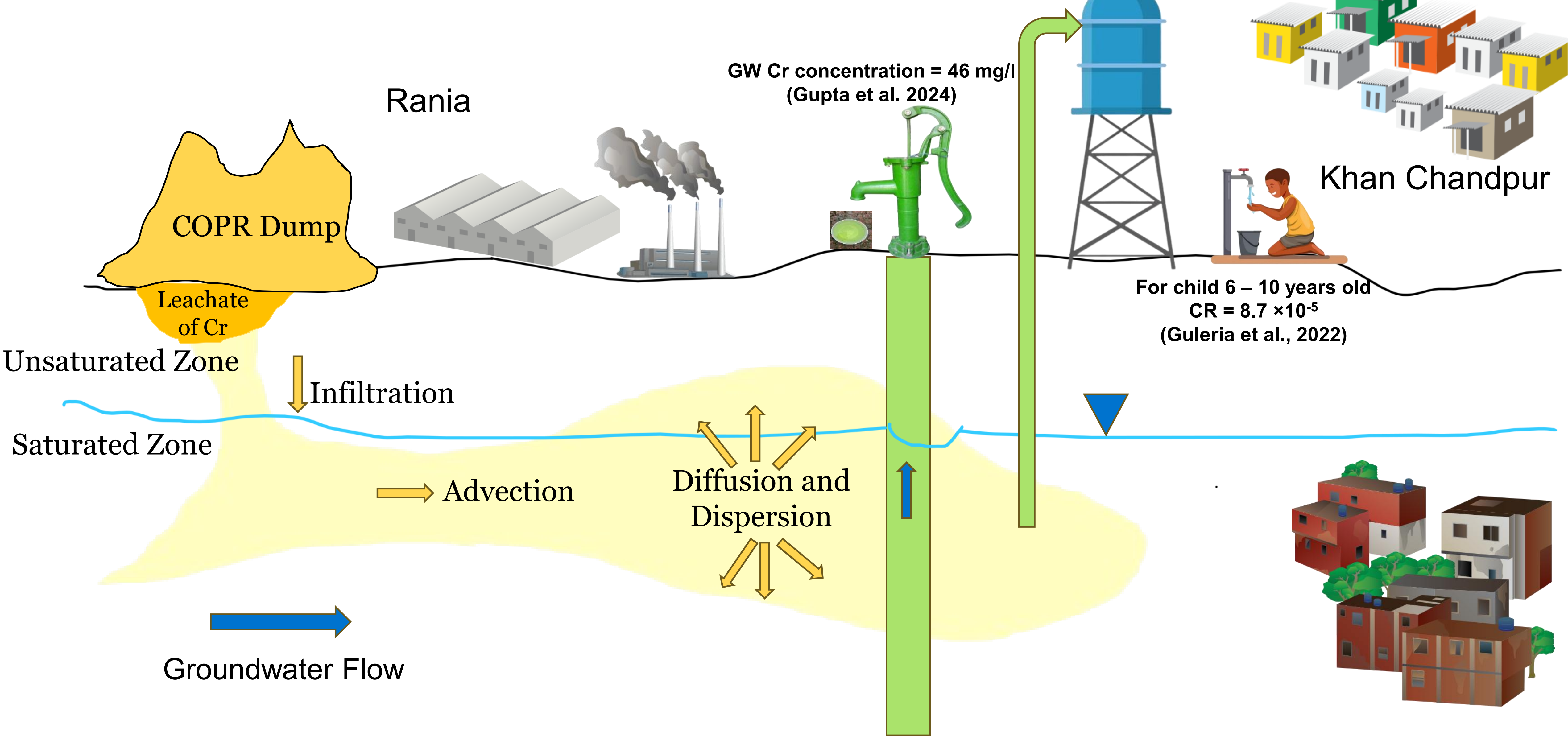
<u>1970-2019:</u> COPR Dumped in Open Area of Rania

*"A better understanding of potential transport pathways, rates, and more efficient clean-up strategies is needed **to improve soil-water (groundwater) quality.**"*

Diagram from E-MicroBiome Project Data Base

LITERATURE REVIEW

Cr-Contamination in Rural Settings



Studies Performed Outside India

Citation	Objective	Study Area	Material	Max Cr concentration
Farmer et al. (2002)	To determine Cr contamination in groundwater and surface water	Glasgow	Groundwater	91 mg L^{-1}
Huang et al. (2009)	To determine the Cr contamination in COPR dump	Hunana, China	Soil	2130.3 mg Kg^{-1}
Driscoll et al. (2010)	Study of Cr contamination in COPR	New Jersey US	COPR	3000 mg Kg^{-1}
Eliopoulos and Megremi (2021)	Status of groundwater for human health	Neogene Assopos and Thiva basins	Soil and Groundwater	40570 mg Kg^{-1} in soil and 21 mg L^{-1} in groundwater
Kazakis et al. 2017)	Cr contamination in groundwater	Sarigkiol Basin of Greece	Groundwater	120 mg L^{-1}

<u>Research Gap:</u> "Only a few Cr-contaminated sites have been investigated till yet over the world. Thus, most of the research questions are yet to be answered."

Studies Performed in India

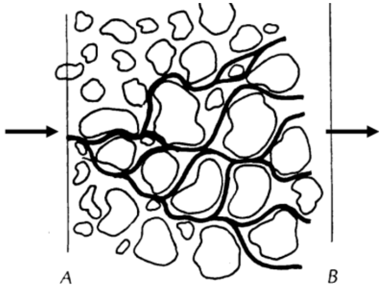
Citation	Objective	Study Area	Material	Max Cr concentration
Kumar et al. (2023)	To determine the Cr in COPR at different depth	Rania, India	COPR	42570 mg Kg^{-1}
Guleria et al. (2022)	To determine the ecological risk, human health risk in groundwater	Rania, Chhiwali and Godhrauli	Groundwater	115 mg L^{-1}
Ratnalu et al. (2022)	To determine the Cr contamination in soil	Peenya, Bengaluru	Soil	718.1 mg Kg^{-1}
Anduyappan et al. (2017)	To determine the Cr concentration in agriculture land	Ranipet, India	Agriculture soil	79.71 mg Kg^{-1}
Krishna et al. (2013)	To determine the heavy metal including Cr in soil	Nuggihalli, Karnataka	Soil	4863 mg Kg^{-1}
Naz et al. (2006)	To determine the human health risk of Cr (VI) and Cr (III) via groundwater.	Sukinda Chromite Mine, Odisha	Groundwater	250.2 mg L^{-1}

<u>Research Gap:</u> "A very little is known about the source, occurrence, fate, transport and remediation of Cr in Indian soil and groundwater system."

Transport Process: Advection-Dispersion

Citation	Study Area	Experimental Conditions and Methods	Dispersion Coff value (D)
Yan et al. (2023)	Shijiazhuang, Zhuzhou and Guangzhou, (China)	Scale: Column experiment	7.84 $cm^2\ h^{-1}$
Trento and Alvarez (2011)	Lab Scale	Scale: Column experiment	10.6 $cm^2\ d^{-1}$
Khan et al. (2010)	Lab Scale	Scale: Column experiment	0.00475 $cm^2\ h^{-1}$
Selim et al. (1989)	Lab Scale	Models: Kinetic model, Convective-dispersive transport equation,	0.0001296 $m^2\ d^{-1}$
Boupha et al. (2004)	Lab Scale	Models: Laplace transforms, The De Hoog algorithm	0.2 $m^2\ d^{-1}$

Dispersion of Cr in soil-water system



$$D\frac{\partial^2 c}{\partial x^2} - v\frac{\partial c}{\partial x} = R_d\frac{\partial c}{\partial t}$$

Where D = Dispersion coefficient, v = groundwater flow velocity of solute, c= concentration, t = time, R_d = Retardation Factor

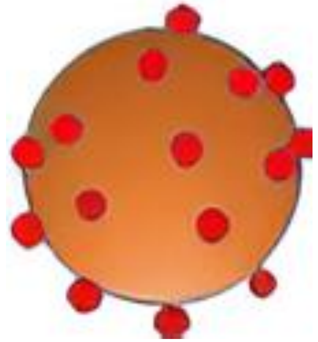
Higher D = larger plume size
Faster v = larger plume size

<u>Research Gap:</u> "Both advection and dispersion parameters are driving variables *which can help villagers to control Cr plume in soil-groundwater systems*. **No site scale data are available.**"

Fate Process: Adsorption

Citation	Study Area	Experimental Conditions and Methods	Adsorption Coefficient K_d ($cm^3\ g^{-1}$)
Ye et al. (2019)	Xiangxiang	Scale: Batch scale and column experiment (40 cm hight and 7 cm diameter)Models: Kinetic models and Isotherm	1.17
Khan et al. (2010)		Scale: Column experiment Analytical Method: UV spectrophotometer	0.0115
Wu et al. (2022)	Xiangxiang City, China	Scale: Column experiment (height 81 cm)	0.0542
Yan et al. (2023)	Shijiazhuang, Zhuzhou and Guangzhou China	Scale: Column experiment (1.2 cm diameter and 8 cm hight) Analytical Method: UV spectrophotometer	0.84
Arthur et al. (2017)	Entisols USA	Scale: Batch adsorption, Models: Freundkich and linear isotherms	0.0051

Adsorption of Cr in soil surface



Equation of Adsorption

$$R_d = 1 + \frac{\rho_b}{\theta} K_d$$

Where R_d is retardation factor, θ is the volumetric water content, ρ_b is the bulk density and K_d is adsorption coefficient.

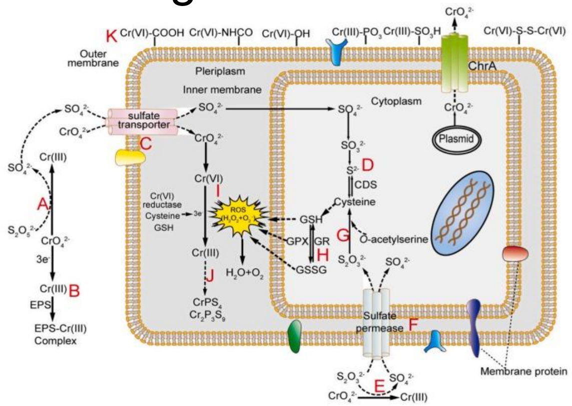
$R_d > 1$; more adsorption
$R_d < 1$; high mobility

<u>Research Gap:</u> "This is very crucial parameter, which required to access Cr in soil-groundwater systems, **yet to be investigated**. *This will help in to select appropriate soil for natural barriers or land reprofiling*."

Fate Process: Reduction, Degradation

References	Plant/Species	Period	Removal efficiency (%)	Medium
Ranieri et al. (2020)	Ailanthus altissima	90 d	56	Soil-groundwater
Liu et al. (2019)	Leersia hexandra Swartz	10 d	100	Soil-groundwater
Chitraprabha and Sathyavathi (2019)	Tagetes erecta	35 d	94	Electropainting effluent
Farid et al. (2017)	Helianthus annuus	56 d	97	Soil
Cheng and Li (2009)	Bacillus	24 h	100	Soil
Maqubool et al. (2015)	Brucella	72 h	92	Waste water
Banerjee et al. (2019)	Bacillus stain	16 h	97	Soil

Removal of Cr from soil-groundwater



Equation

$$CrO_4^{2-} + 8H^+ + 3e^- \rightarrow Cr^{3+}\ 4H_2O$$

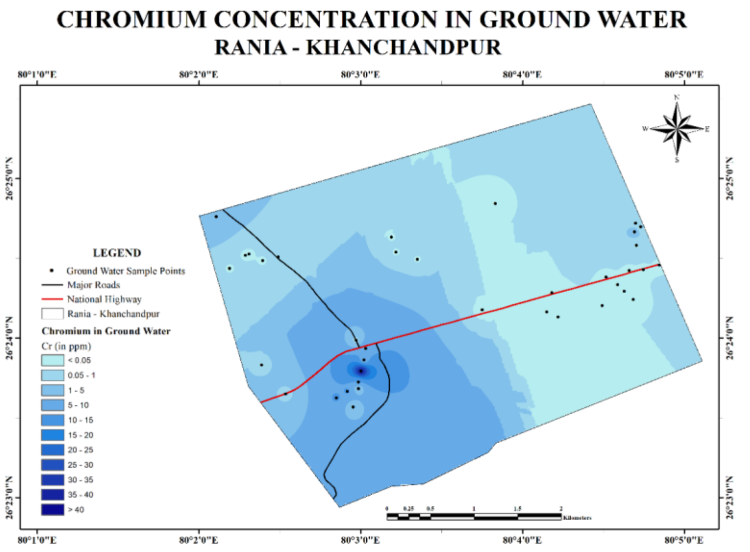
$$\frac{-\partial C}{\partial t} = k_1 C \qquad \text{where } k_1 = \text{Decay Coefficient}$$

<u>Research Gap:</u> “A very limited small-scale experiments performed earlier however remediation of Cr in soil-groundwater system is crucial for safe drinking water production.”

Studies Performed in Rania and Other Villages

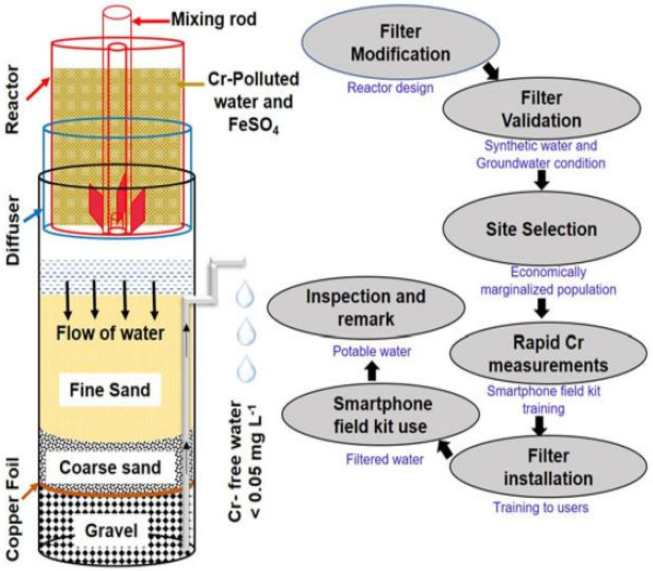
Groundwater Assessment

Gupta et al. (2024), Kumar et al. (2023), Guleria et al. (2022), Martein et al. (2017), Foldi et al. (2013)



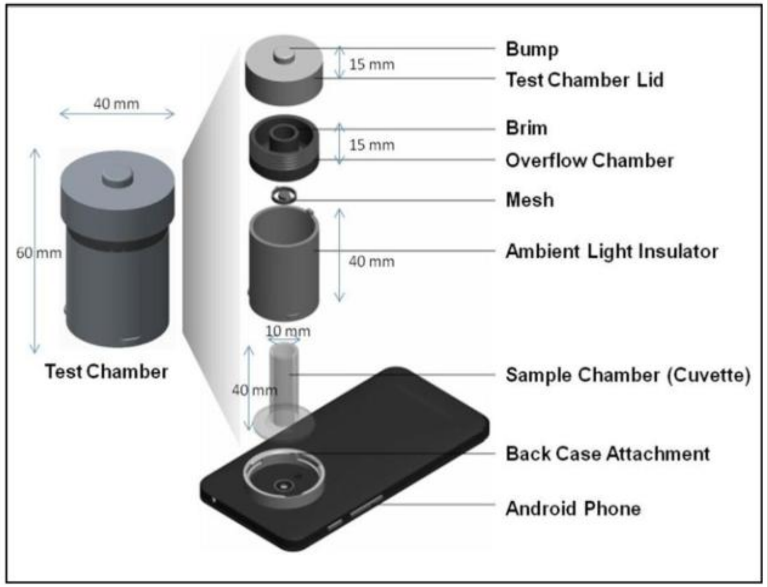
Filtration System

Singh et al. (2023), Bandyopadhyay et al. (2023)



Smartphone-based Monitoring Tool

Singh et al. (2021)



<u>Research Gap</u>: "**The in-depth investigation** of the soil and groundwater system in the Rania-Khan Chandpur sites **is at an initial stage**."

Research Gaps

- Although Gupta et al. (2024), Kumar et al. (2023), etc. have performed the soil-groundwater quality assessment, **the fate and transport of Cr in soil-groundwater systems in Rania and Khan Chandpur villages has not been investigated yet**. <u>The same is the condition in other contaminated villages in India</u>.

- Despite the available studies, **limited studies have been performed on a large/village scale to understand the contamination issues, improve community participation, develop remediation/restoration designs**, etc.

- Considering the seriousness of this site, **remediation efforts for Cr-contaminated soil-water systems in Rania-Khan Chandpur have not been initiated**, as reported by the Central Pollution Control Board (CPCB, 2023), and Hon’ble NGT.

- A limited effort has been made to provide information to the residents or to receive their feedbacks to improve the soil-water management practices.

Objectives

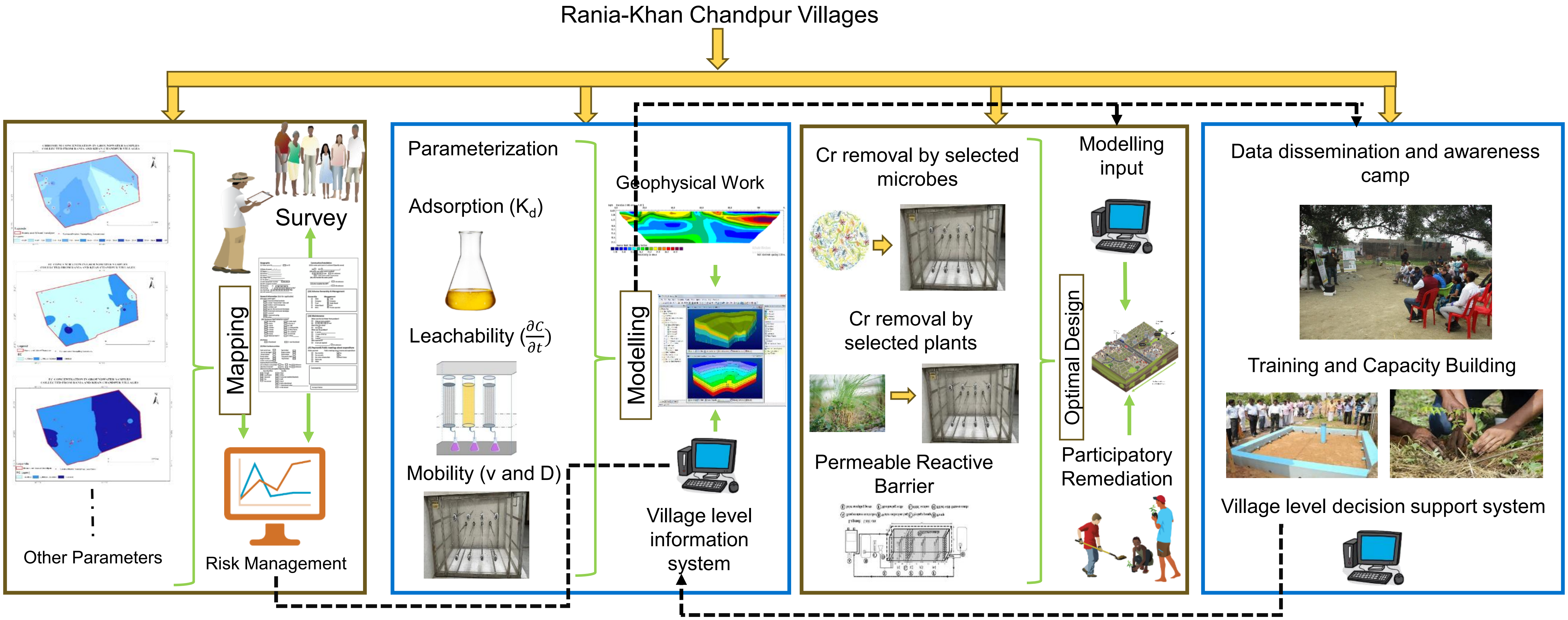
- To conduct soil and groundwater quality assessment and questionaries-based survey to analyze the Cr-contamination issues and its awareness among local people in Rania and Khan Chandpur villages.

- To investigate the fate and transport of Cr in soil-groundwater systems in and around Rania-Khan Chandpur villages.

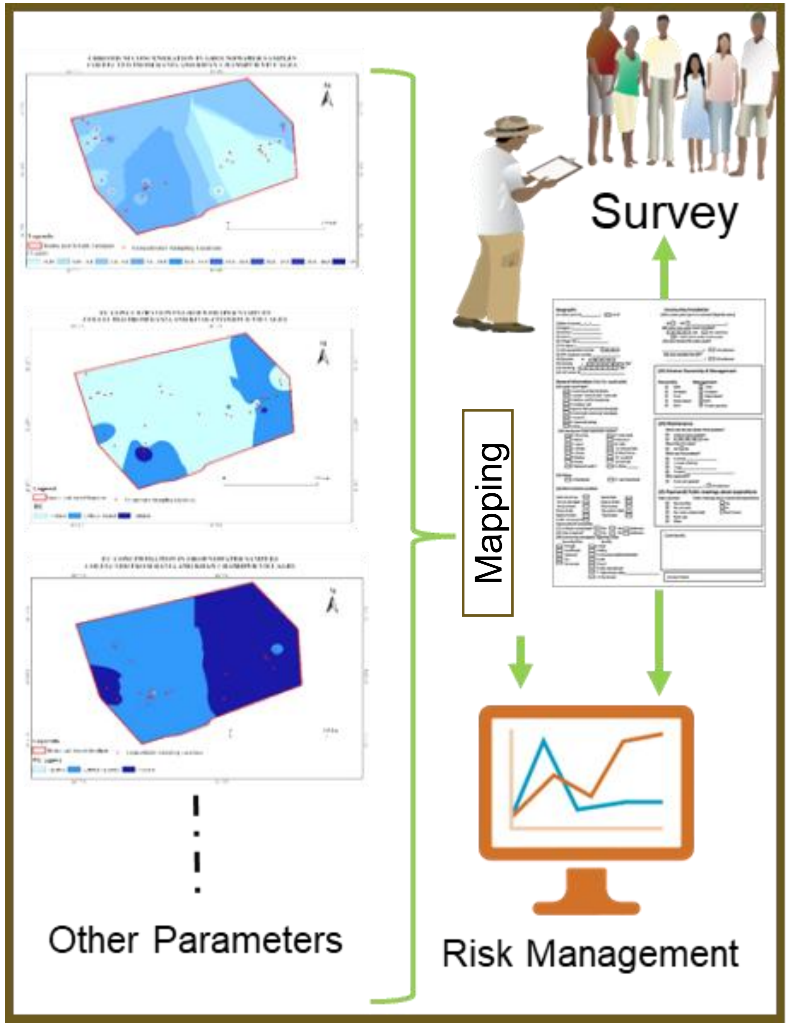
- To design remediation techniques for Cr contaminated soil-groundwater systems and demonstrate the optimal remediation design in Rania and Khan Chandpur villages.

- To develop capacity-building programs through village level intervention for a better community participation in the soil-groundwater resources management in both the villages.

Overall Work Plan



Methodology for Objective 1



- Sampling (seasonal) of soil (n=70, from 0-150 cm bgl), surface water (n=55), and groundwater samples (165 samples).
- All the samples are analyzed as per **IS: 3025 (1984).**
- ICP-MS analysis and all parameter analyses performed as recommended by CPCB.
- Field-scale monitoring of groundwater table and other hydrometeorological variables.
- Questionnaire-based survey (one-to-one survey, family level survey, stakeholders survey, gram panchayat) in Rania-Khan Chandpur villages (n=as much as possible).
- Risk communication and training on soil-groundwater quality analysis. (Part of Objective 4).

Output: **This work will guide villagers in installing bore wells in safe zones**, reducing the risk of Cr-contamination, and establishing a village-level soil-groundwater information system.

Methodology for Objective 2

Laboratory Experimentation

Numerical Modelling

Task 1: Parameterization: Estimation of adsorption coefficient


pH: 5; Water content: 80%, 100%, 120%


pH: 7; Water content: 80%, 100%, 120%


pH: 9; Water content: 80%, 100%, 120%

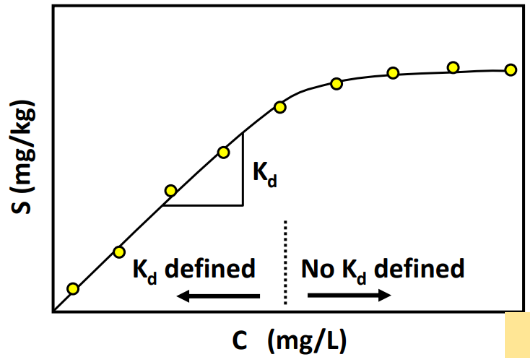

pH: 11; Water content: 80%, 100%, 120%

Pseudo-first-order equation

$$\frac{dq_t}{dt} = k_1(q_e - q_t)$$

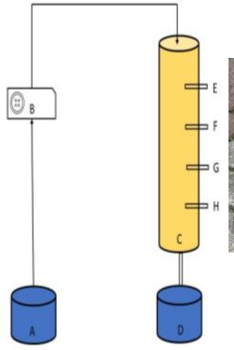
Langmuir equation

$$\frac{C_e}{q_e} = \frac{C_e}{q_{max}} + \frac{1}{bq_{max}}$$



K_d value

Task 2: Parameterization: Estimation of leachability index and dissolution



- Estimation of dissolution coefficients for Cr from COPR core (n=3)
- Estimation of dissolution coefficients for Cr from soil core (n=3)

Fick's Law

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Optimization of parameters using HYDRUS 1D

$$D_L = \alpha_L v + D^*$$

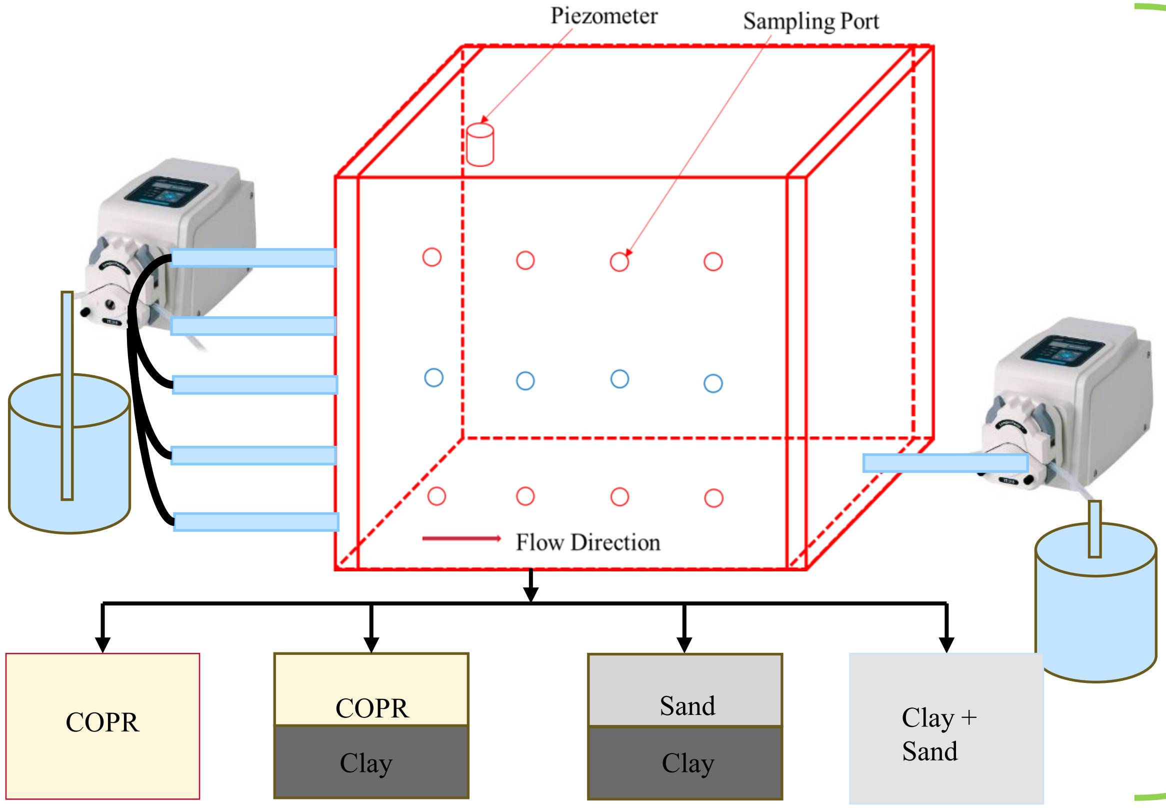
Upscaling to 3D flow system

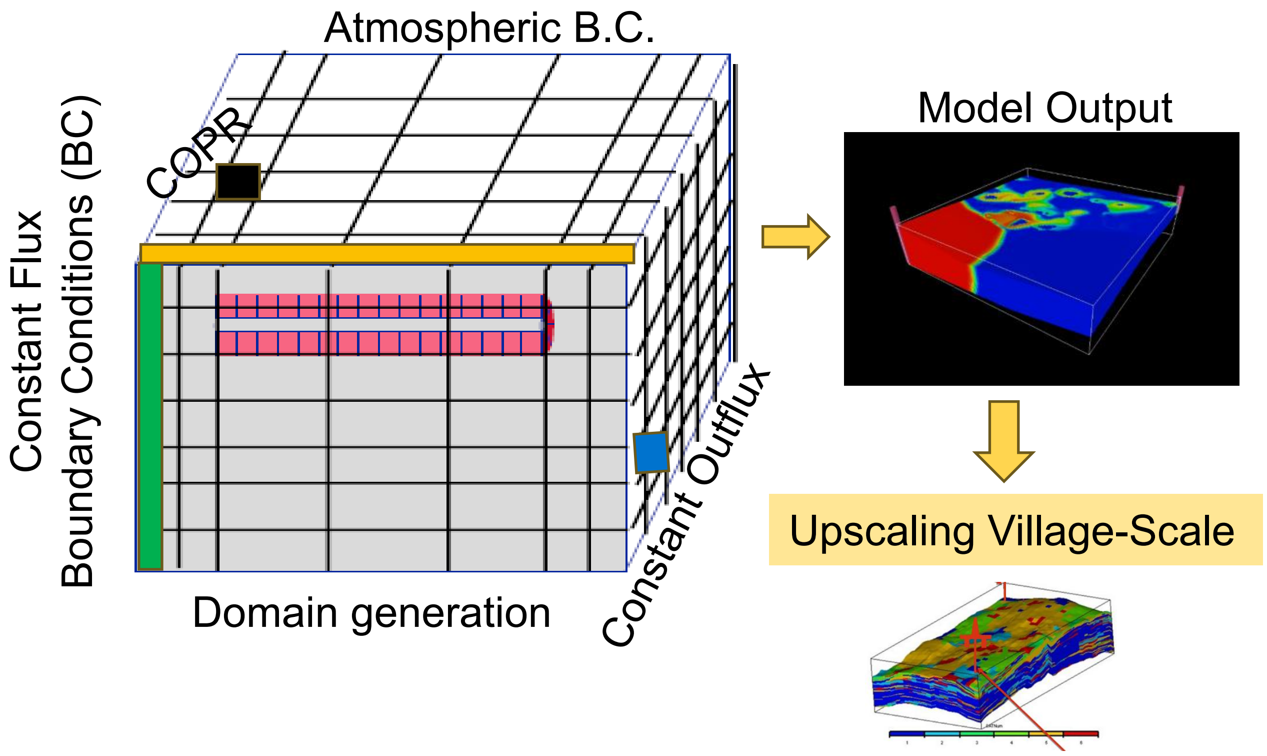
<u>Output:</u> This study will assist villagers in choosing suitable soils for chromium retardation, as well as for building road foundations, houses, and other structures.

Methodology for Objective 2 (Cont….)

Laboratory Experimentation | *Numerical Modelling*

Task 3: Upscaling Cr-fate and transport variables to 3D scale





<u>Output:</u> Again, this study will assist villagers in choosing suitable soils for chromium retardation, as well as for building road foundations, houses, and other structures.

Methodology for Objective 2 (Cont….)

Task 3A: Fabrication of 3D Agar & Core collection from Rania and Khan Chandpur Villages, *Method adopted from Previati et. al., (2018)*

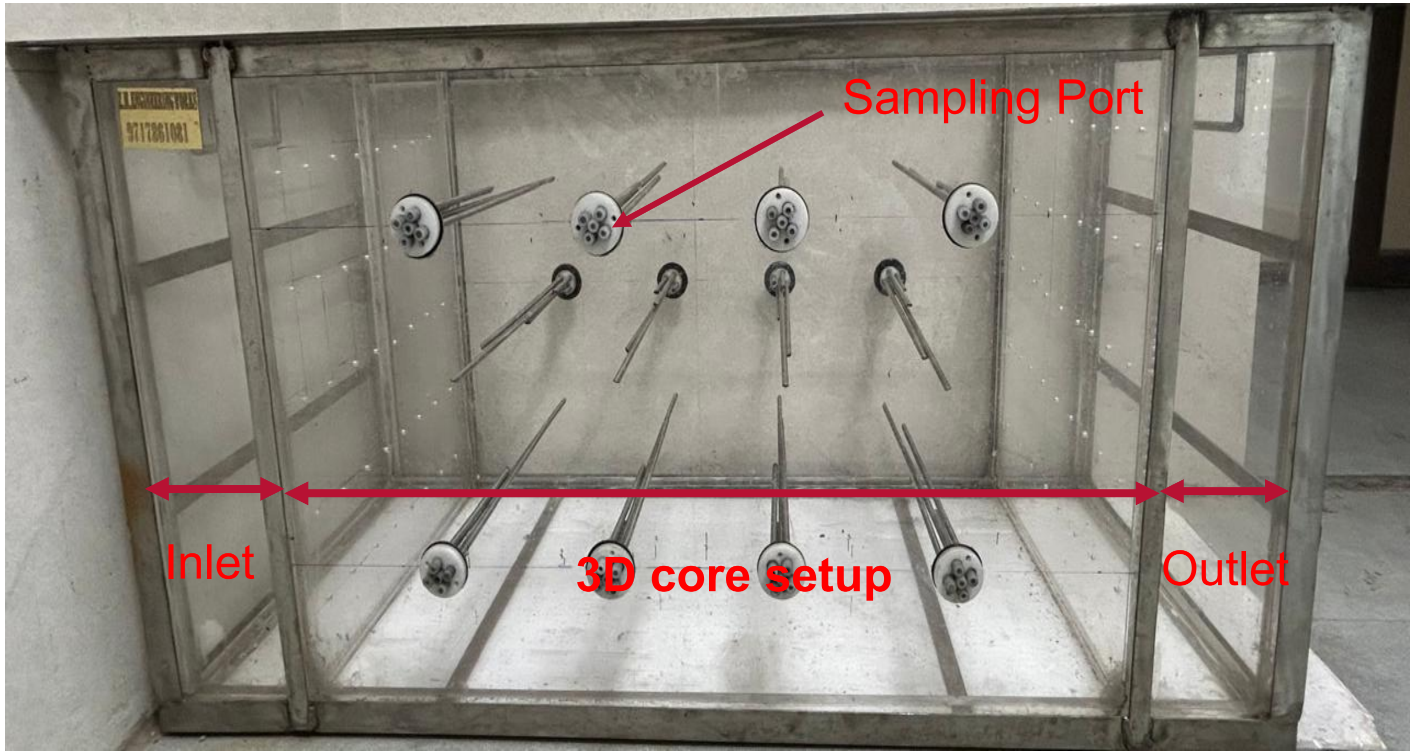
3D Agar system has been fabricated

- The core will be collected from nearby the Rania COPR dump site. The size of the core collected in the field will be 100 × 50 × 65 cm.
- The core will be set by cutting the edges to the tank size and place on the tank.

Output: This is a unique setup, which will help us to collect undisturbed cores for large-scale experiments.

Methodology for Objective 2 (Cont….)

Task 3B: Fabrication of 3D Sand Tank, ***Method adopted from Gupta and Yadav, (2020)***



Dimensions of the 3D tank: 90 × 45 × 60 cm
Sampling ports: 12

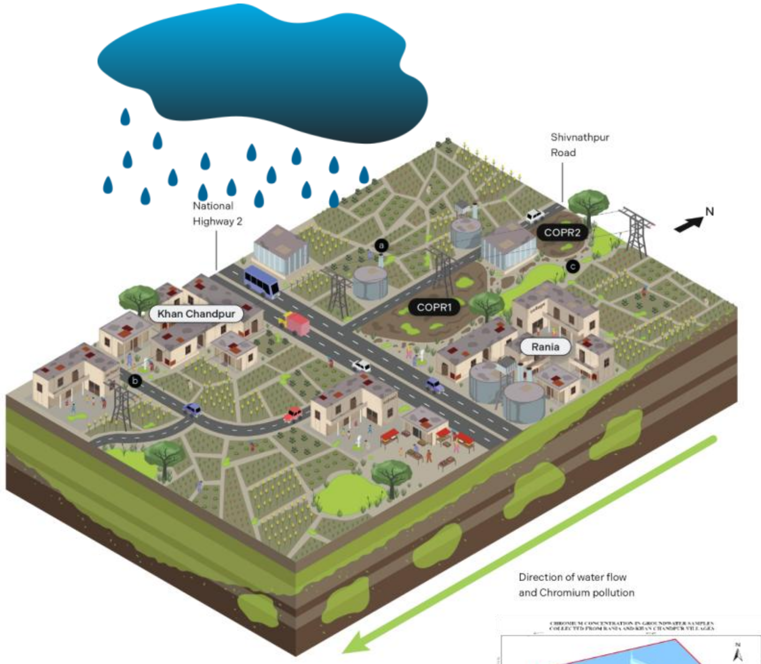
Front view of 3D tank for transport study of Cr in soil-groundwater system

<u>Output:</u> This setup will help us to estimate the water flow and Cr transport parameters for upscaling its mobility at the village level.

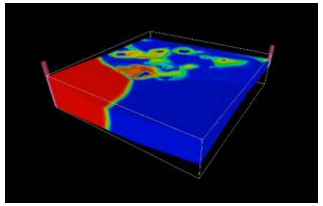
Methodology for Objective 2 (Cont….)

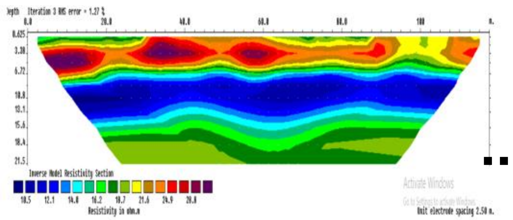
Task 4: Cr-mobility simulation at village-level

Obj. 1: Hydrology: IMD Data for present and CNRM-CM6 Data for Future study

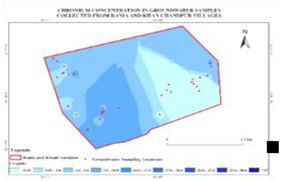


Obj.2, Task 1-3: Water and Solute: VGM Parameters

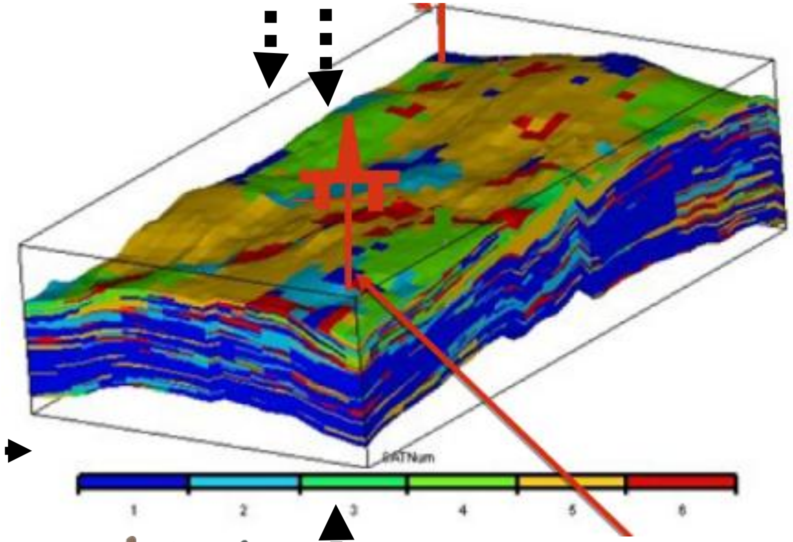




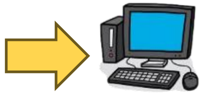
From E-Microbiome Project: Hydrogeology: Geophysical Data



Obj. 1: Soil-Water Quality Data



Village level Soil-Groundwater information system



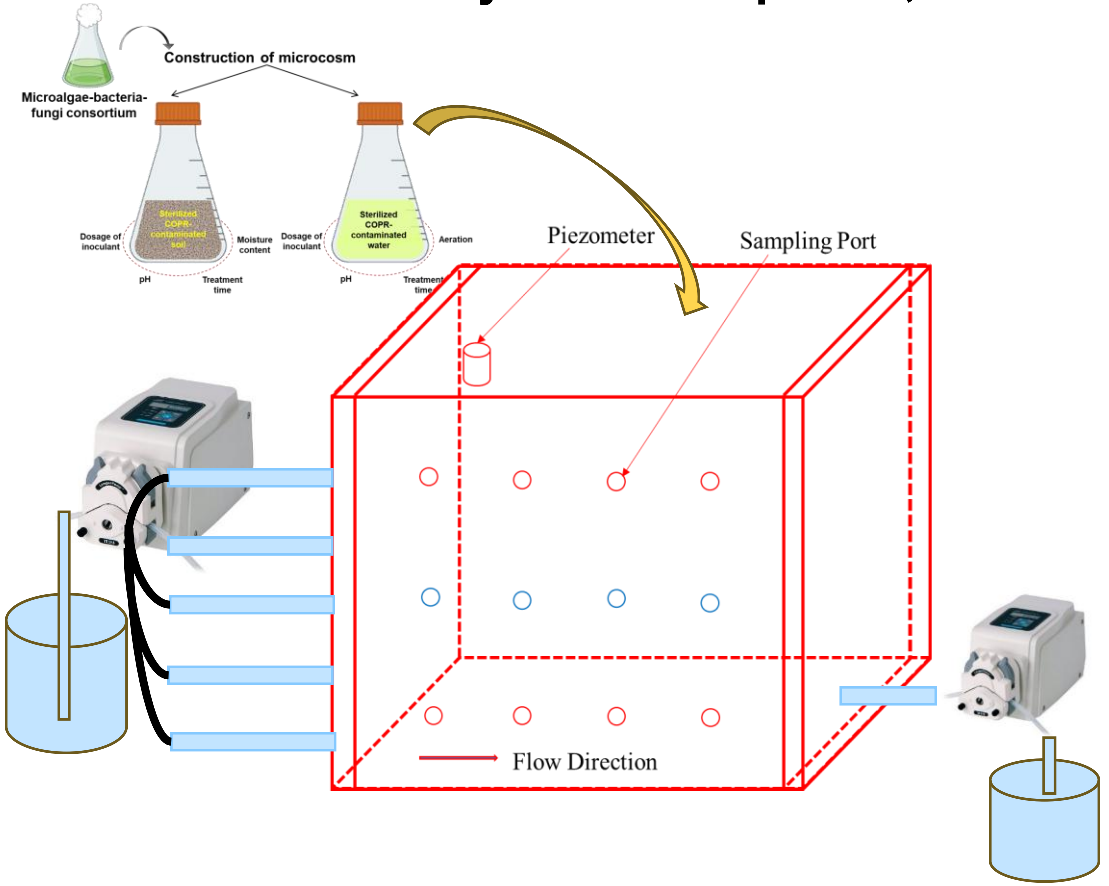
Obj. 1: Public Feedback

Output: This study will assist villagers in soil-groundwater management, like installation of bore wells, areas for remediations, control of pumping, etc.

Methodology for Objective 3

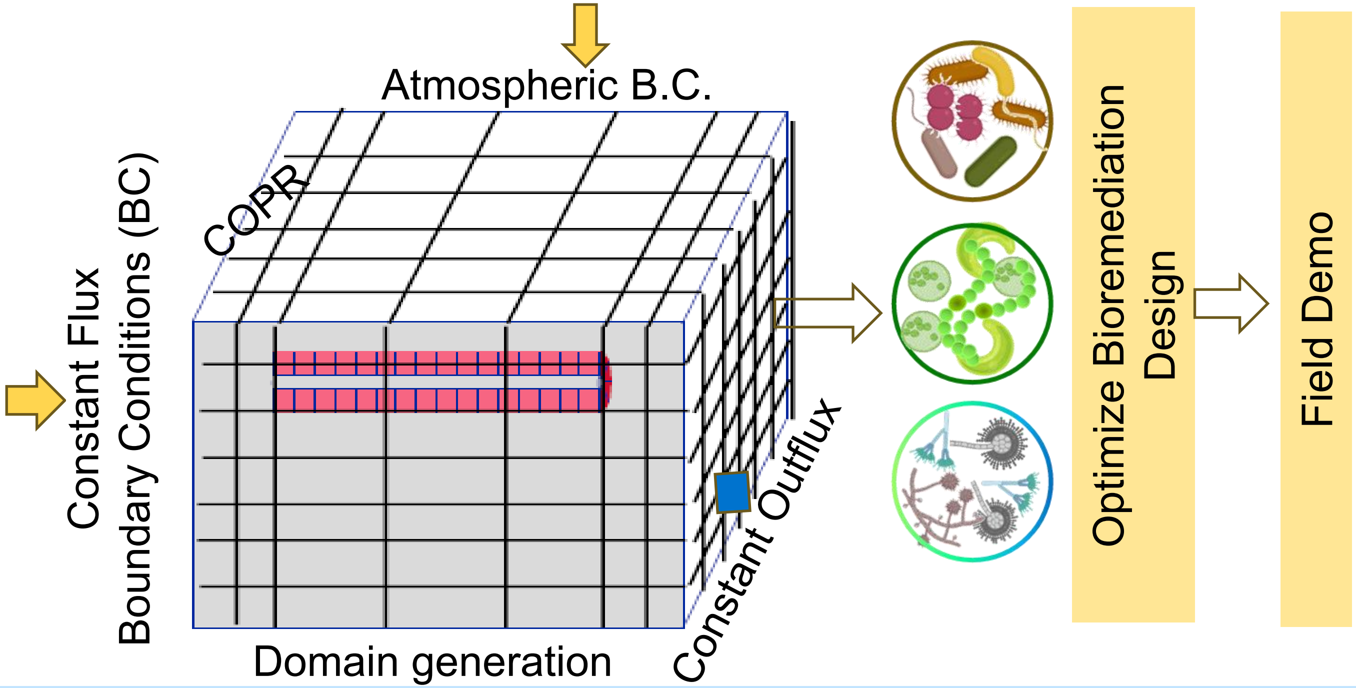
Laboratory Experimentation

Task 1: Cr removal by bacterial species, consortium, etc.



Numerical Modelling

Kinetics model	Condition	Equation	Rate constant
Constant or zero-order	$X_0>>C_0$; $C_0>>K_S$	$-\partial C/\partial t=k_0$	$k_0=\mu_{max}X_0$
Linear or first-order	$X_0>>C_0$; $K_S>>C_0$	$-\partial C/\partial t=k_1C$	$k_1=\mu_{max}X_0/K_S$
Monod or Michaelis–Menten	$X_0>>C_0$	$-\partial C/\partial t=k_mC/(K_S+C)$	$k_m=\mu_{max}X_0$
Logistic	$K_S>>C_0$	$-\partial C/\partial t=k_lC(C_0+X_0-C)$	$k_l=X_0/K_S$
Logarithmic	$K_S>>C_0$	$-\partial C/\partial t=k(C_0+X_0-C)$	$k=\mu_{max}$



<u>Output:</u> This experiment will help practicener to select appropriate microorganism to initiate community led bioremediation.

Methodology for Objective 3 (Cont….)

Field Demonstration

Numerical Modelling

Task 2: Cr removal by plants.

Layout of Field Experiment

$$V^{rs}\frac{\partial C^{rs}}{\partial t} = Q_{trans}C^{w} + \left(C^{w} - \frac{C^{rs}}{P_C}\right)D_r V^{rs} - \frac{V_{max}C^{rs}}{K_m + C^{rs}}\,M_r$$

V_{max} is the maximum uptake rate of the metal by roots from the root surface (mg of metal per kg weight of root biomass per day),
K_m is the half saturation constant (solution concentration when uptake is half maximal and
M_r is the root biomass

Optimize Phytoremediation Design

Field Demo

<u>Output:</u> This experiment will help practicener to select appropriate plant species to initiate community led bioremediation.

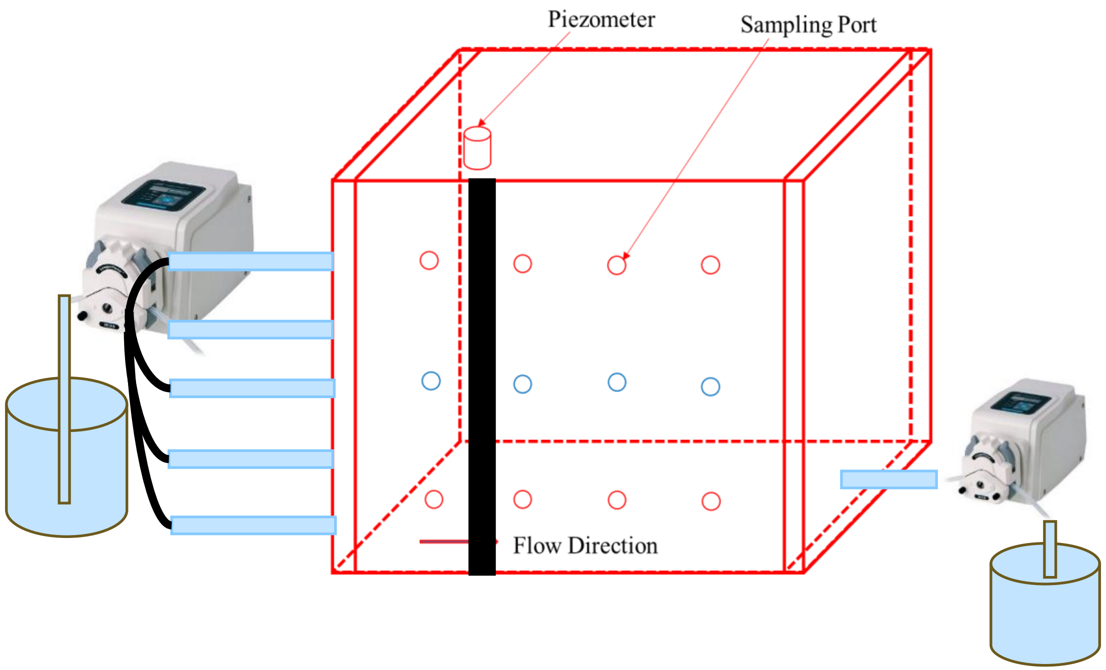
Methodology for Objective 3 (Cont….)

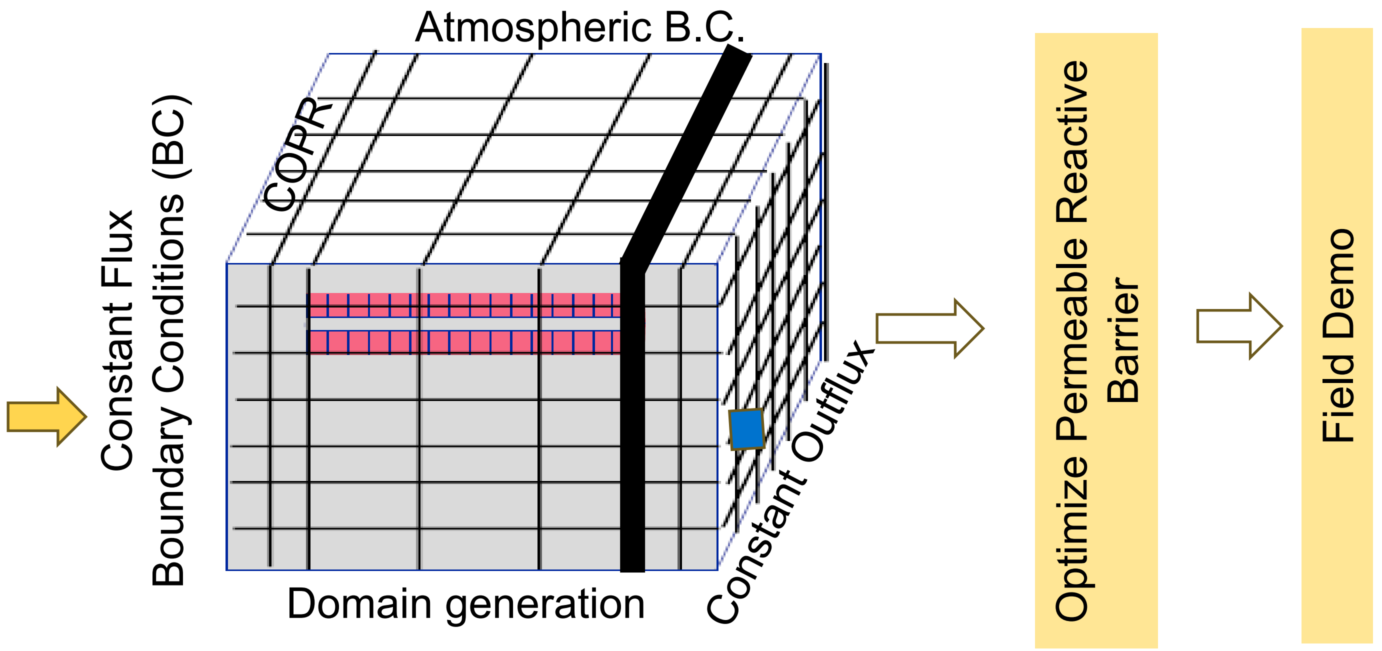
Laboratory Experimentation | *Numerical Modelling*

Task 3: Cr Removal by Permeable Reactive Barrier

Material: Biochar
Dimensions: 10 × 45 ×60 cm

Parameters to Optimize: Thickness of PRB, Groundwater Velocity, Cr concentration





<u>Output:</u> This experiment will help practicener to create the barrier to control Cr plume in the field at Rania and Khan Chandpur Villages.

Methodology for Objective 4

Task 1: Data Dissemination and Awareness Camp

Target: Residents

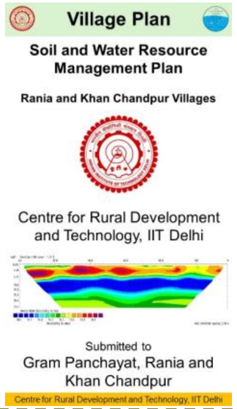
Task 2: Train the Trainers (Practiceners)

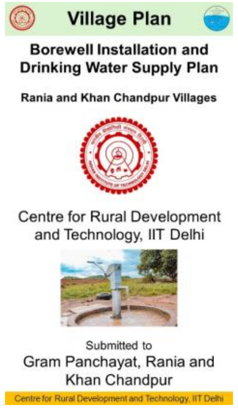
Target groups: People Representatives, Persons involved in bore wells digging professions, PHED officials, etc.)

- Appropriate plant selection
- Appropriate location for borewell
- Appropriate borewells for drinking water
- Appropriate design for borewells

Task 3: Develop Village Soil-Water Information Systems

Target groups: Community





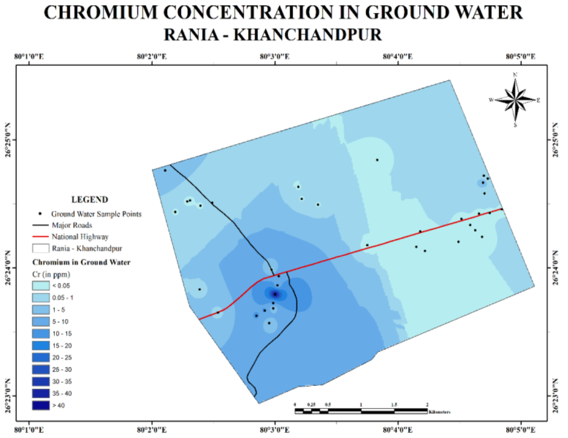
Task 4: Creating a pool of Jalrakshak (Volunteers)

Target groups: Community

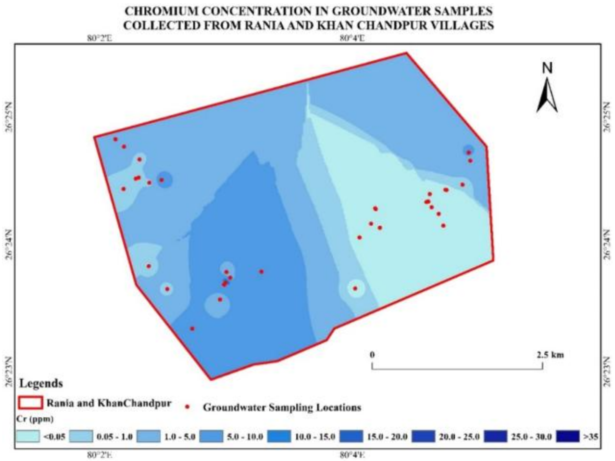
WORK DONE

Groundwater Quality Assessment

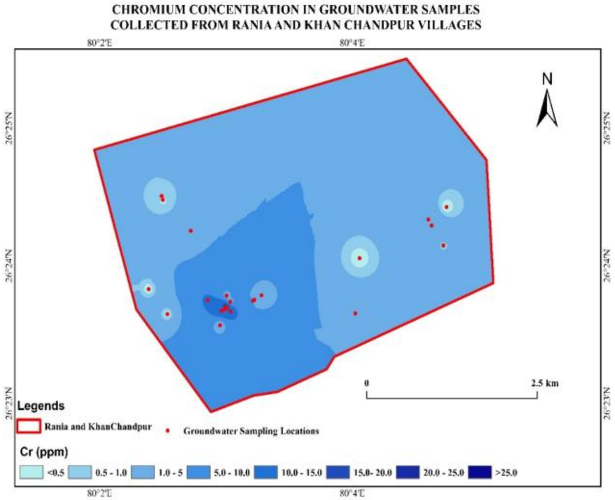
- The concentration of Cr is much higher than the permissible drinking limit (**0.05 mg L^{-1}**) in Rania-Khan Chandpur.
- As compared to **Rania (0 – 10.45 mg L^{-1})** Cr concentration is higher in the **Khan-Chandpur Village (0 – 46 mg L^{-1})** .



Cr -Contamination in October 2022*



Cr-Contamination in June 2023



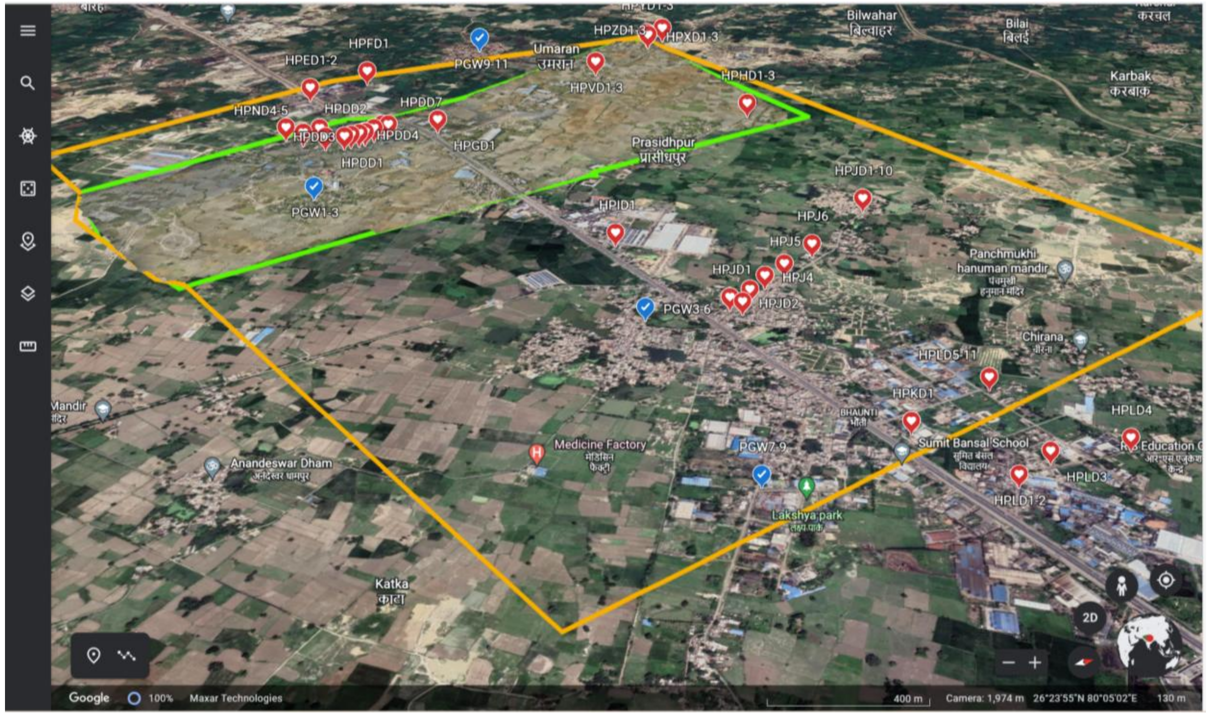
Cr-Contamination in February 2024

When COPR was there

When COPR was removed

Risk Communication

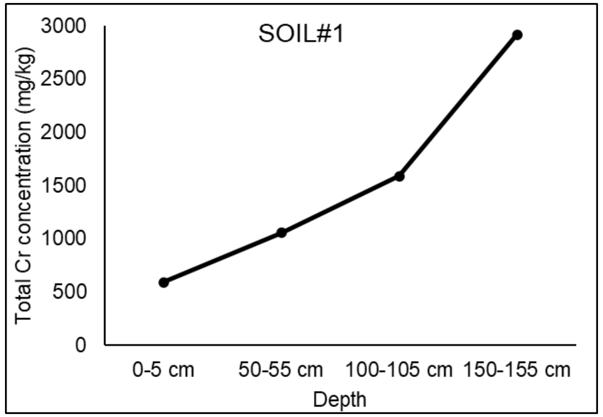
[Communication] *[Indicators]* *[Google Maps, Social Media, Mobile Applications]*

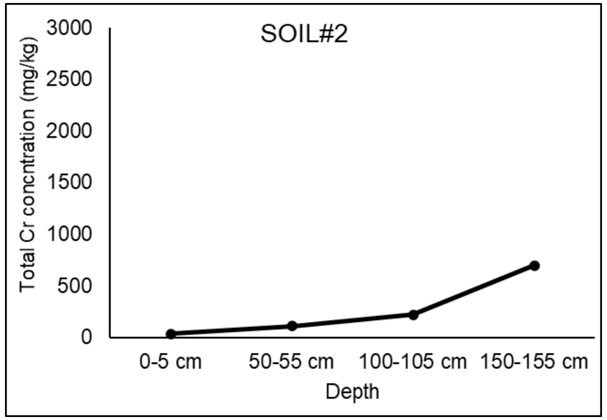


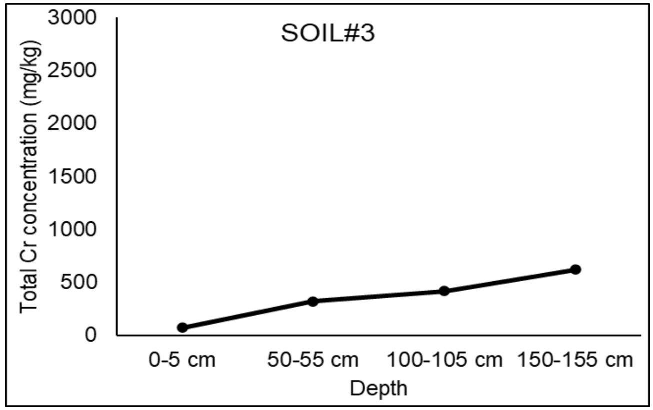
Output: We are providing appropriate assistant to the residents so that they can avoids such highly contaminated drinking water.

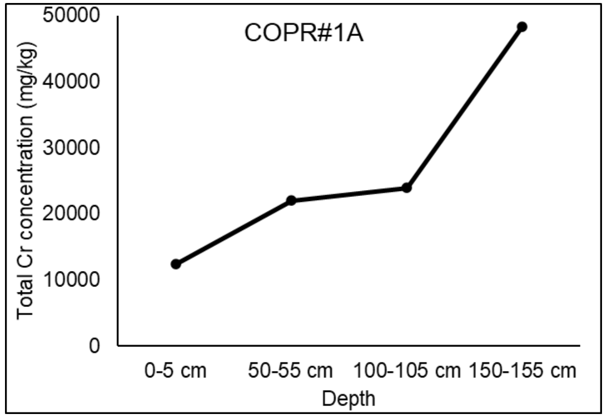
Soil Quality Assessment

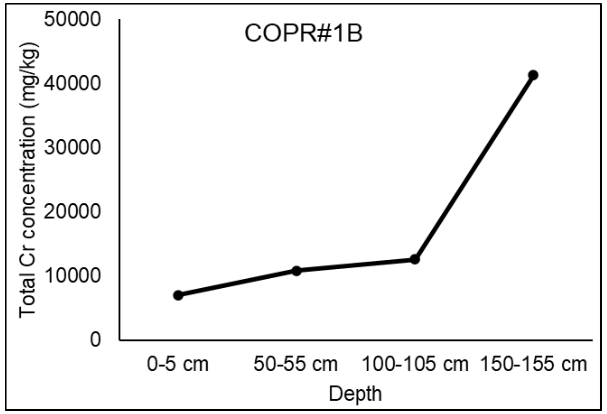
We have performed Cr analysis for soil and COPR cores collected from Rania- Khan Chandpur.

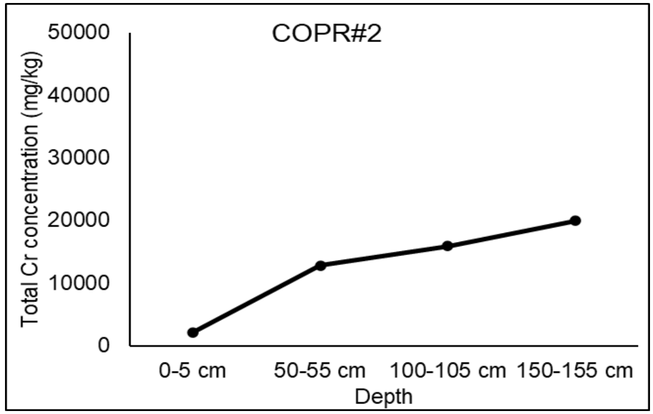












<u>Output:</u> Before our presence in Rania, residence were frequently using COPR and COPR mix soil in house and road foundations etc. We suggested to avoid any such black, brown and green soils/COPR.

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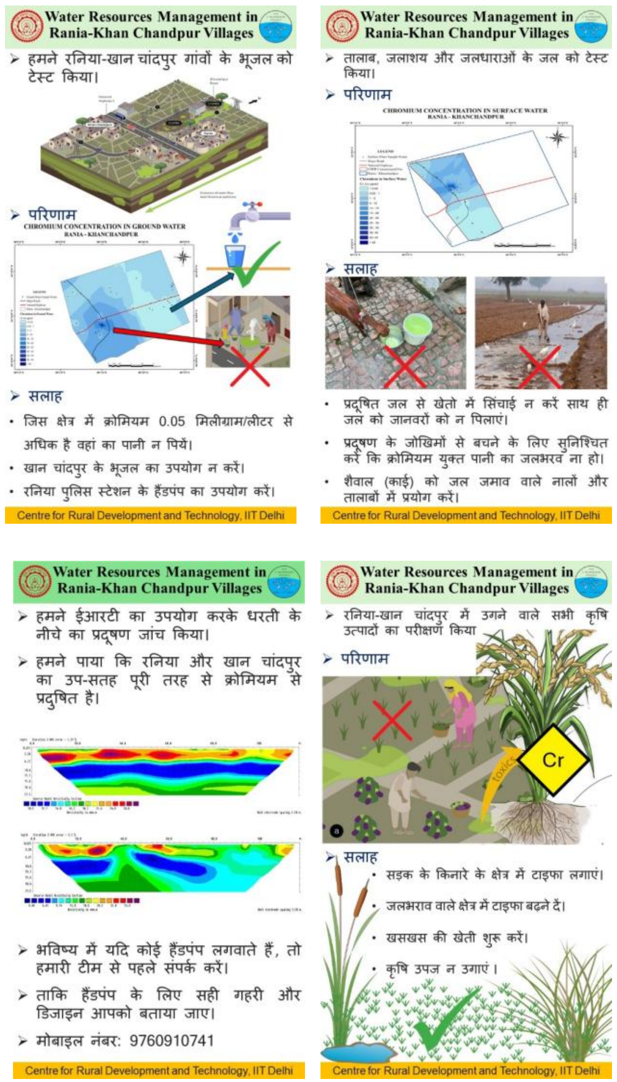

Data Dissemination and Awareness Camp

1st Prize Winner of 16th Open House at IIT Delhi

Donated Prize Amount

Key Learnings:

1. Community Water Sensitivity was limited.
2. Pumping is driving variable need to be considered.
3. Decentralized water management can be achieved through bio/phytoremediation.

3D Core Collection

Problem Faced (During Soil Core Collection)

Core cutting by JCB was unsuccessful.

Next Steps:

1. **Manual Core Cutting: Proceed with manual cutting for precision.**
2. **Alternative Approach: JCB can use to cut on two sides, then manually finish the process with labors.**

3D Tank Experiment

Problem: Soil insert in sampling rod.

Solution: Create a hole using another rod of the same diameter.

Insert the rod into the newly created hole.

THANK YOU